

Advanced (Habanero)**Q1.** What is the expansion of $(x + y)^2$?

- (A) $x^2 + 2xy + y^2$
 (B) $x^2 - 2xy + y^2$
 (C) $x^2 + xy + y^2$
 (D) $x^2 - xy + y^2$
 (E) $x^2 + y^2$

Q2. What is the factored form of $x^2 + 6x + 9$?

- (A) $(x + 3)^2$
 (B) $(x - 3)^2$
 (C) $(x + 3)(x - 3)$
 (D) $(x + 2)(x + 4)$
 (E) $(x - 2)(x - 4)$

Q3. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
 (B) $x^2 - 4x + 4$
 (C) $x^2 + 4x - 4$
 (D) $x^2 - 4x - 4$
 (E) $x^2 + 2x + 1$

Q4. What is the expansion of $(x - y)^2$?

- (A) $x^2 + 2xy + y^2$
 (B) $x^2 - 2xy + y^2$
 (C) $x^2 + xy + y^2$
 (D) $x^2 - xy + y^2$
 (E) $x^2 + y^2$

Q5. What is the factored form of $x^2 - 8x + 16$?

- (A) $(x + 4)^2$
 (B) $(x - 4)^2$
 (C) $(x + 2)(x + 6)$
 (D) $(x - 2)(x - 6)$
 (E) $(x + 3)(x - 5)$

Q6. Which of the following is not a perfect square trinomial?

- (A) $x^2 + 6x + 9$
 (B) $x^2 - 6x + 9$
 (C) $x^2 + 6x - 9$
 (D) $x^2 - 6x - 9$
 (E) $x^2 + 2x + 1$

Q7. What is the expansion of $(2x + 3y)^2$?

- (A) $4x^2 + 12xy + 9y^2$
 (B) $4x^2 - 12xy + 9y^2$
 (C) $4x^2 + 6xy + 9y^2$
 (D) $4x^2 - 6xy + 9y^2$
 (E) $4x^2 + 9y^2$

Q8. What is the factored form of $x^2 + 10x + 25$?

- (A) $(x + 5)^2$
 (B) $(x - 5)^2$
 (C) $(x + 2)(x + 8)$
 (D) $(x - 2)(x - 8)$
 (E) $(x + 3)(x + 7)$

Open Ended Questions (FRQ)**Q9.** Prove that $(x + y)^2 = x^2 + 2xy + y^2$ using the distributive property.**Q10.** Factor the expression $x^2 + 12x + 36$ and show that it is a perfect square trinomial.**Chef's Tips**

- Provide clear and concise instructions to students.
- Make sure students have the necessary tools and equipment.
- Encourage students to ask questions and seek help when needed.